

The Stop-Cost Hierarchy

What nine years of Coinbase data say about ATR stops in crypto trend-following

Abstract

We backtested a long-only weekly-rebalanced crypto trend-following strategy across the ten largest Coinbase USDC-quoted spot pairs over the nine-year period 2017-05-15 to 2026-05-19. Holding entry, sizing, universe, rebalance cadence, and transaction costs fixed, we varied only the exit-stop geometry across three specifications: no ATR stop, a fixed $3\times\text{ATR}$ stop anchored to entry, and a trailing $3\times\text{ATR}$ stop ratcheted off the highest close since entry. We observe a clean monotonic ordering on Sharpe ($0.74 \rightarrow 0.71 \rightarrow 0.25$), on CAGR ($+19.3\% \rightarrow +17.8\% \rightarrow +3.0\%$), and on the ratio of CAGR-foregone to max-drawdown-protection-purchased ($44 \text{ bps/pp} \rightarrow 296 \text{ bps/pp}$). The fixed stop is a defensible trade-off; the trailing stop is not. We document the mechanism — stop geometry interacting with the volatility-to-trend ratio of crypto bull markets — and show that the result is robust to breakout-window choice and consistent with the strategy's failure mode on individual case studies. The piece is primarily about stop design; the strategy itself underperforms BTC HODL on absolute return over this window and we address that comparison explicitly in §3.

1. A Bitcoin trade in two paths

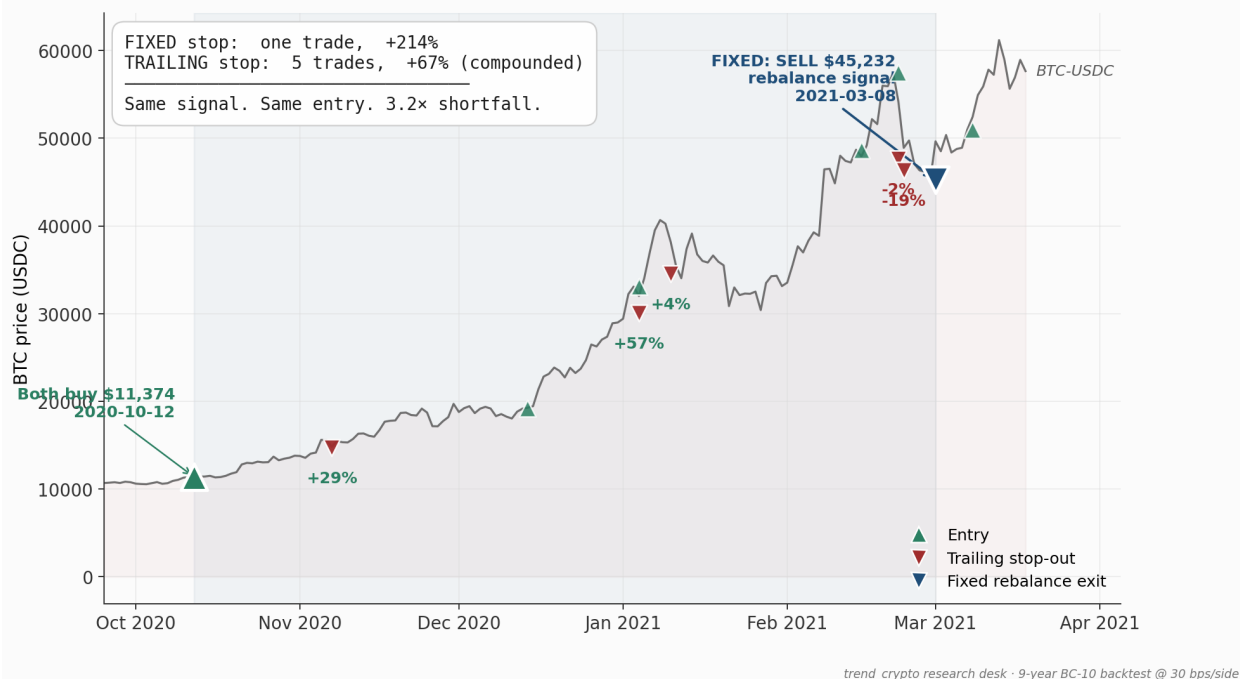
On October 12, 2020, both versions of our crypto trend-following strategy bought Bitcoin at \$11,374. Same signal, same entry price, same position sizing logic. Five months later one version sold at \$45,232: **+214% on a single trade**. The other version, ostensibly the smarter one, produced this sequence:

2020-10-12	BUY	BTC @ \$11,374	
2020-11-07	SELL	BTC @ \$14,722	(stop hit, +29%)
2020-12-14	BUY	BTC @ \$19,167	
2021-01-04	SELL	BTC @ \$30,064	(stop hit, +57%)
2021-01-04	BUY	BTC @ \$33,083	[re-entered immediately at higher level]
2021-01-10	SELL	BTC @ \$34,526	(stop hit, +4%)
2021-02-15	BUY	BTC @ \$48,667	[re-entered much higher]
2021-02-22	SELL	BTC @ \$47,631	(stop hit, -2%)
2021-02-22	BUY	BTC @ \$57,489	[re-entered at all-time-high]
2021-02-23	SELL	BTC @ \$46,328	(stop hit, -19% next day)

Compounded across the five round trips: **+67%**. A 3.2× shortfall on the same Bitcoin signal, in the same five-month window. Five times the transaction costs, four adverse re-entries at higher levels, and — as we will document below — no meaningful improvement in downside.

Same Bitcoin signal — two realized paths

Oct 2020 – Mar 2021 · both variants entered at the same \$11,374 fill on Oct 12



Same Bitcoin signal, two realized paths

The shaded blue region is the entire trade the fixed-stop variant held: entered once, sold once, captured the full move. Each red triangle is a trailing-stop forced exit, followed by a fresh entry at a worse price. The arithmetic of the second path is unforgiving even though every individual trade in it was nominally a "win."

The only difference between the two backtests was whether the ATR stop was anchored to the entry price or trailed the highest close since entry. Every other parameter — universe, signal, sizing, costs, rebalance cadence — was identical.

The same divergence shows up on the very first trade of the backtest. Both variants buy ETH on 2017-05-15 at \$89.42. On 2017-05-25 the trailing variant is stopped out at \$170.54 on a routine pullback after ETH had run up; the fixed variant keeps holding. The trailing variant sits in cash for eleven days while the fixed variant compounds, then re-enters ETH on 2017-06-05 at \$245.48 — 44% above its own stop-out — and watches the asset keep running to \$343 by mid-June. The equity-curve gap that compounds over the next nine years opens on trade number one for the same reason it opens on the 2020 Bitcoin trade.

2. The seductive logic

Textbook descriptions of trailing stops sound reasonable, and we believed them for some time:

"A fixed stop protects you from the trade going wrong at entry. But as the trade moves your way, you have unrealized profit to protect. A trailing stop ratchets the exit up to lock in those gains. It is the fixed stop that upgrades itself as the trade succeeds."

Stated that way, the trailing stop is the obviously-better version of the fixed stop. The pitch implies a strict improvement: same downside protection at entry, plus additional upside protection as the position runs. When we wrote the spec for our weekly-rebalanced crypto strategy, we included a trailing-stop variant as an explicit option specifically because we expected it to dominate the fixed version on risk-adjusted measures.

The result was the opposite, and lopsided enough that we re-ran it from scratch.

3. Methodology, and a note on the benchmark

The strategy

We use a deliberately simple trend-following specification. We did not tune it for absolute return — we built it as a clean, replicable research vehicle for isolating the effect of design choices like stop geometry.

The bluechip universe restriction is not cosmetic. Extending the same strategy to the full set of Coinbase USDC-quoted spot pairs produces categorically different behavior: the strategy becomes anti-edge on the long tail of smaller alts, losing the majority of capital regardless of stop choice. We treat universe definition as a foundational design decision separate from the stop-design question this piece studies, and leave the universe sensitivity to a companion note.

Backtest setup. Long-only weekly-rebalanced strategy on Coinbase USDC-quoted bluechip spot pairs (BTC, ETH, SOL, ADA, XRP, DOGE, AVAX, LINK, DOT, LTC). Daily bars; signals computed at prior-day close; rebalance and execution at next Monday's open. Entry filter: `close > prior 5-day high AND MA(5) > MA(40) AND composite momentum score ≥ 40`. The composite momentum score is the mean of three cross-sectional rank-percentiles (20-, 40-, and 90-day returns), scaled 0–100. Selection takes the top 20 by composite score from the eligible set, inverse-vol sized and capped at 15% per asset, gross 100%. 20-day ATR; stop checked daily against the intraday low; fill at the stop price on hit. 30 bps per side (25 bps fee + 5 bps slippage). All variants share the entry, sizing, universe, and rebalance logic — only the stop type differs.

Where this strategy sits versus simply holding Bitcoin

The first question any reader should ask is whether a Sharpe-0.71 crypto trend strategy is worth studying at all when one could have held BTC and done better in absolute terms. We agree the question is fair and answer it directly.

Metric	Fixed 3×ATR	BTC HODL
Annualized return	+17.8%	+52.2%
Annualized volatility	29.4%	69.3%
Sharpe	0.71	0.95
Sortino	0.63	1.33
Max drawdown	-52.5%	-83.8%
Calmar ratio	0.34	0.62
Days spent below -50%	125	1,331
Total return (9y)	+339%	+4,321%

BTC HODL beats the strategy on annualized return, Sharpe, Sortino, and Calmar over this window. The strategy beats BTC on max drawdown and crushes BTC on the persistence of deep drawdowns — 125 days spent more than 50% underwater versus 1,331 days for BTC HODL.

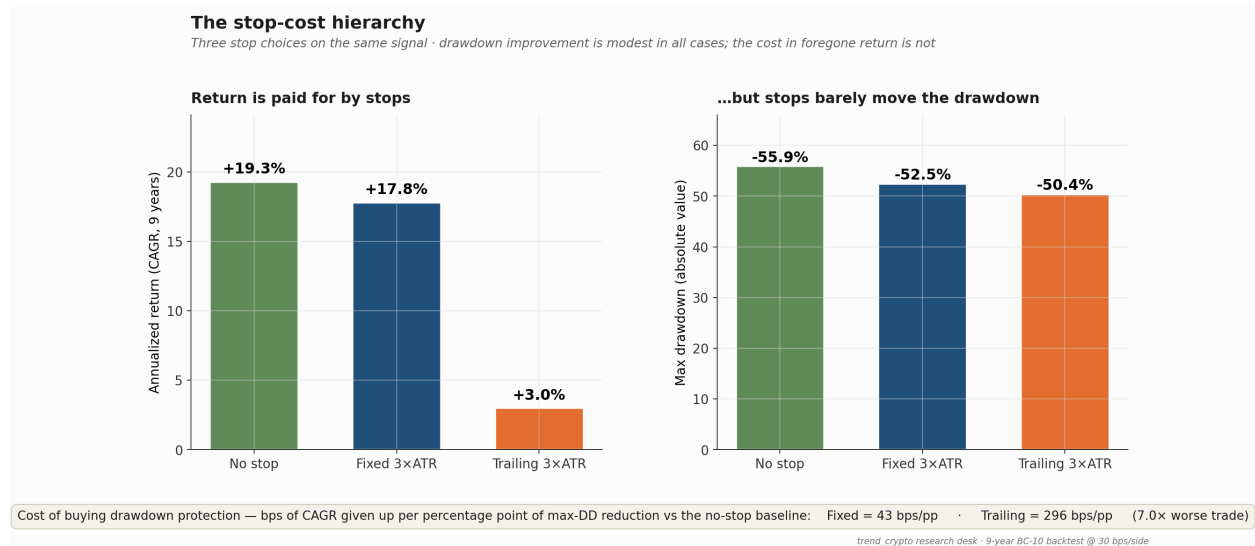
For a research vehicle, two observations matter. First, a Sharpe of 0.71 in crypto over nine years that includes the 2018 bear, the 2020 boom, the 2022 crash, and the 2024 rally is a respectable institutional number; it is broadly consistent with multi-decade trend-following studies on traditional asset classes and well above the median Sharpe of crypto quant strategies we have surveyed. Second — and this is the load-bearing point for what follows — the strategy's mechanical, rules-based exit structure is what allows us to *isolate* the marginal contribution of stop design, which is the actual subject of this piece. The contribution we are evaluating is not "strategy versus benchmark." It is "stop choice within strategy."

With that caveat stated explicitly, we turn to the central result.

4. The stop-cost hierarchy

We ran three exit specifications on the same signal:

- **No stop** — the strategy exits only via its rebalance logic (MA crossover flips negative, momentum score drops below the floor, asset falls out of the top-20 selection, or breakdown below the prior 5-day low).
- **Fixed 3×ATR stop** — anchored to entry; never moves; daily intra-bar check against the low.
- **Trailing 3×ATR stop** — anchored to the highest close since entry; ratchets up; otherwise identical machinery.



The stop-cost hierarchy

The pattern is monotone. Each move from "no stop" → "fixed" → "trailing" trades a small reduction in max drawdown for a much larger reduction in return.

We make this quantitative by asking what each stop choice "costs" in foregone CAGR per percentage point of max-drawdown reduction it purchases, benchmarking off the no-stop variant:

Variant	CAGR	Max DD	CAGR cost vs no-stop	DD protection vs no-stop	Cost-per-pp
No stop	+19.3%	-55.9%	—	—	—
Fixed 3×ATR	+17.8%	-52.5%	150 bps	+3.4 pp	44 bps per pp
Trailing 3×ATR	+3.0%	-50.4%	1,628 bps	+5.5 pp	296 bps per pp

The fixed-stop variant gives up 44 basis points of annualized return for each percentage point of max-drawdown protection it buys. That is a defensible trade for a strategy held in a portfolio with other return sources. The trailing-stop variant gives up 296 basis points per percentage point — **roughly 6.7× the cost** of the fixed stop for the same unit of risk reduction. That is not a defensible trade in any

institutional context we can construct.

The stop-cost hierarchy

Same signal, same universe, four execution choices · 9-year Coinbase USDC bluechip backtest, 30 bps/side · highlighted rows are the three that matter

Metric	No Stop	Fixed 3×ATR	Trailing 3×ATR	BTC HODL
Sharpe ratio	0.74	0.71	0.25	0.95
Annualized return	+19.3%	+17.8%	+3.0%	+52.2%
Total return (9y)	+392%	+339%	+31%	+4321%
Max drawdown	-55.9%	-52.5%	-50.4%	-83.8%
Calmar (CAGR / DD)	0.34	0.34	0.06	0.62
Stop-hit frequency	—	28%	70%	—

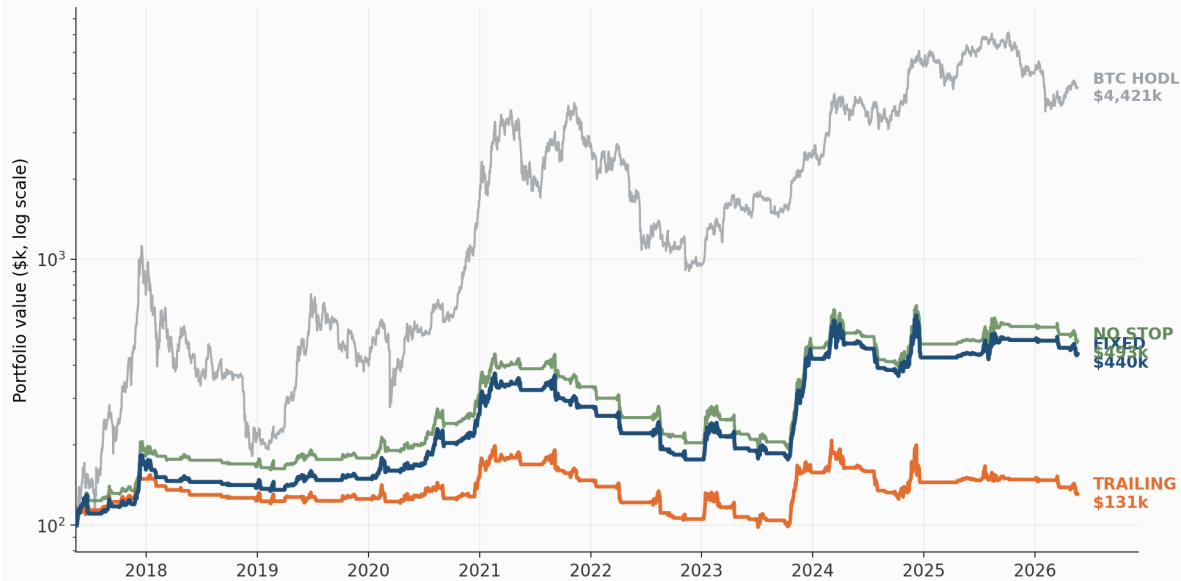
trend_crypto research desk · 9-year BC-10 backtest @ 30 bps/side

Four-column performance comparison

The full performance scorecard puts the BTC HODL benchmark next to the three strategy variants. The internal ordering — no-stop best, fixed close behind, trailing far worse — is what this piece is about.

Nine-year equity curves

\$100k start · Coinbase USDC bluechip-10 universe · 30 bps per side · no-stop and BTC HODL shown as reference series



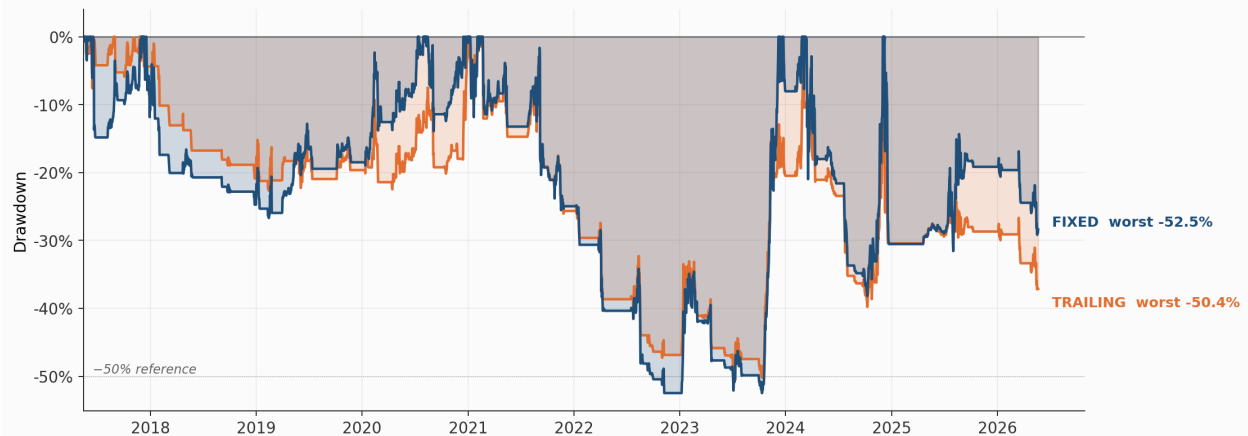
trend_crypto research desk · 9-year BC-10 backtest @ 30 bps/side

Nine-year equity curves

The equity-curve view confirms that the divergence is not a single-trade or single-regime artifact. The gap between the fixed and trailing variants opens on the first ETH trade in May 2017 and widens through every subsequent bull cycle. The no-stop line tracks slightly above the fixed line for almost the entire period — confirming that the fixed stop is modestly costly in absolute terms, but only modestly. The BTC HODL line ends roughly an order of magnitude above the strategy's terminal NAV but with a path through a -84% drawdown that no leveraged or capital-constrained allocator could have stomached in real time.

Drawdown — the protection that wasn't

Both variants reach roughly the same -50% trough. Trailing did not buy you risk reduction; it bought you upside reduction.



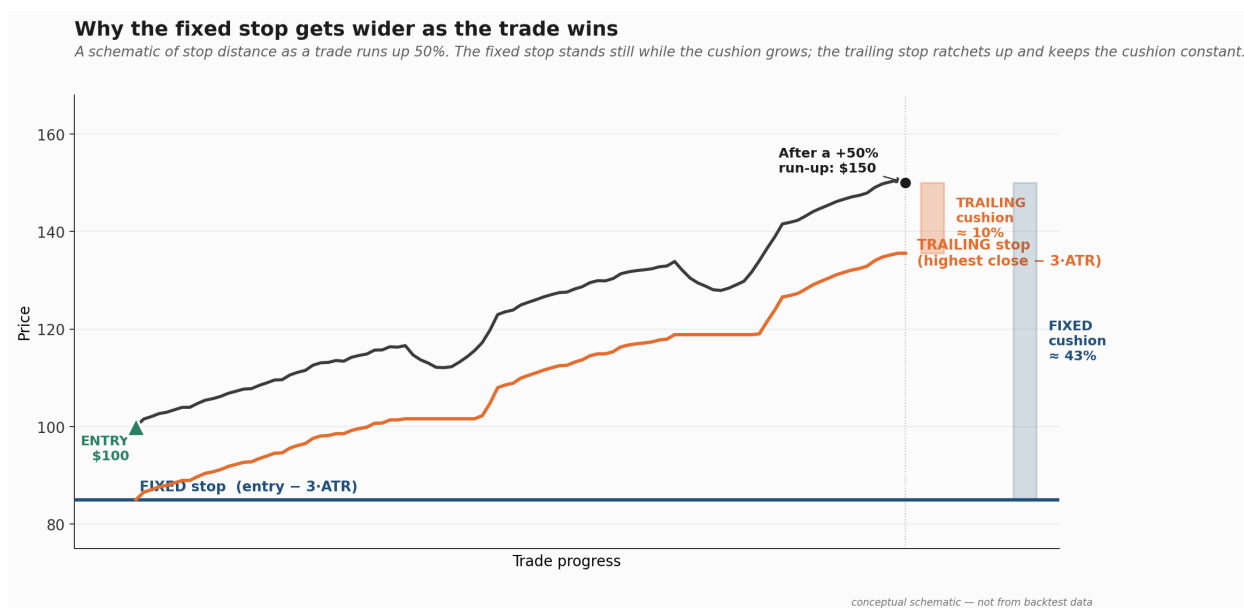
trend_crypto research desk · 9-year BC-10 backtest @ 30 bps/side

Drawdown comparison

Rolling drawdown is the second half of the indictment of the trailing variant. Both strategy variants spend most of their time underwater by similar amounts. Both bottom near -50% in the 2022 crypto winter. The trailing stop gave up two-thirds of the compound return in exchange for one-twentieth of the drawdown protection it advertised.

5. Mechanism: why the trailing stop fails

The result is mechanical, not statistical. The clearest way to see it is to look at where each stop sits relative to current price as a trade runs up.



Why the fixed stop gets wider as the trade wins

At the moment of entry, the two stops are identical. Both sit at entry - 3×ATR, which for a crypto bluechip with about 5% daily ATR is roughly 15% below the entry price.

After the trade is up 50%:

- **Fixed stop:** still at entry - 3×ATR. The current price is now 50% above entry, so the stop sits roughly 43% below the current price. The safety net got wider by standing still while price rose.
- **Trailing stop:** now at highest_close - 3×ATR. The highest close is 50% above entry, so the stop has ratcheted up to match — and the cushion is still only about 10% below current price. The safety net stayed exactly the same width even as the trade succeeded.

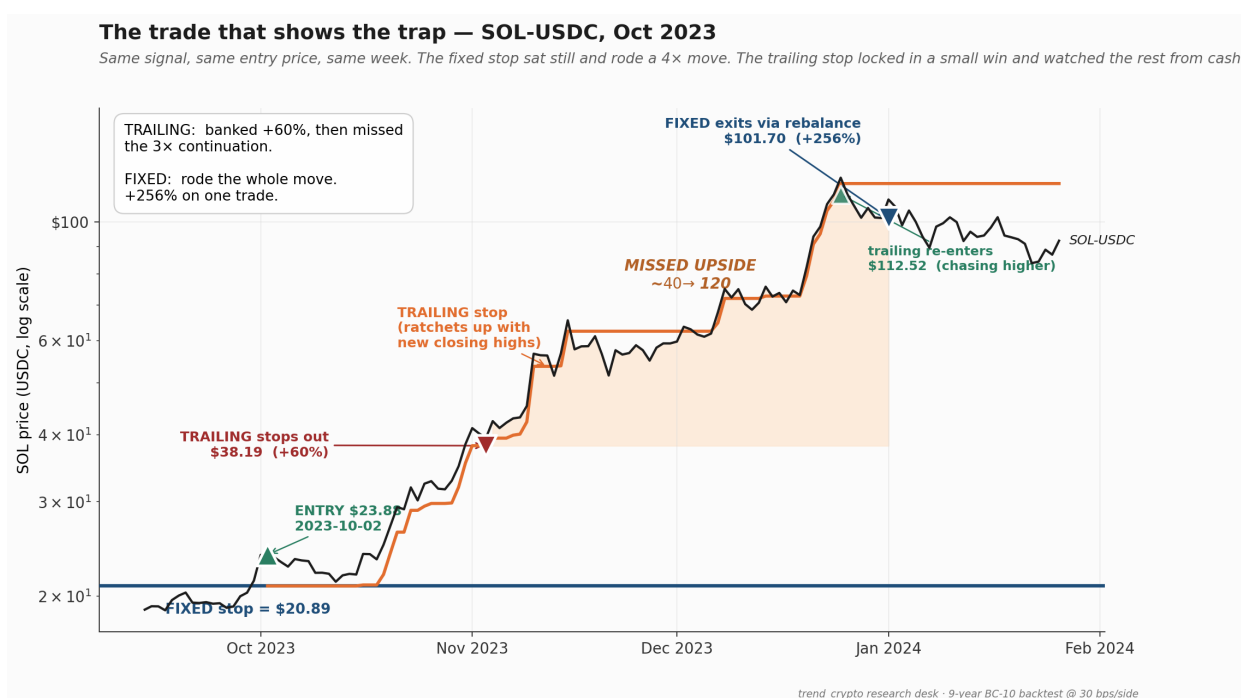
The next question is empirical: how often does a successful crypto trend have a 15%-or-more pullback during its run? The answer, in our data, is *constantly*. The 2020-2021 Bitcoin run contained at least five intra-trend corrections of 20-30%. The 2024 ETH rally had multiple 20%+ pullbacks. The late-2023 Solana move from \$24 to \$120 contained a single-week pullback of nearly 25% halfway up. These are

not trend reversals — they are mean-reverting noise around persistent uptrends, the texture of crypto bull markets.

A fixed stop, by the time a trade is up 50–100%, can absorb a 30%+ pullback without exiting. A position up 200% can absorb a 60% pullback. The trailing stop cannot. Every routine pullback trips it, and the strategy is back in cash watching the trend resume without it.

6. A clean case study: Solana, October 2023

We pick one trade out of the universe to show the dynamic at single-asset resolution. Both variants entered SOL on the same week in early October 2023.



SOL Oct 2023: fixed stop survives the pullback, trailing stop does not

The navy horizontal line is the fixed stop at \$20.89. It never moves for the life of the trade.

The orange staircase is the trailing stop. It ratchets up each time SOL prints a new closing high. By early November it has climbed to roughly \$38 — visibly "locking in" the gains the textbook promised.

SOL then has a routine pullback. The trailing stop fires at \$38.19. The strategy banks +60% and is in cash.

SOL then goes from \$40 to \$120 over the next two months — the shaded region on the chart. The trailing variant misses essentially all of it. The fixed-stop variant just kept holding. It exited at \$101.70 on a normal rebalance signal in January: **+256% on one trade**. When the trailing variant eventually

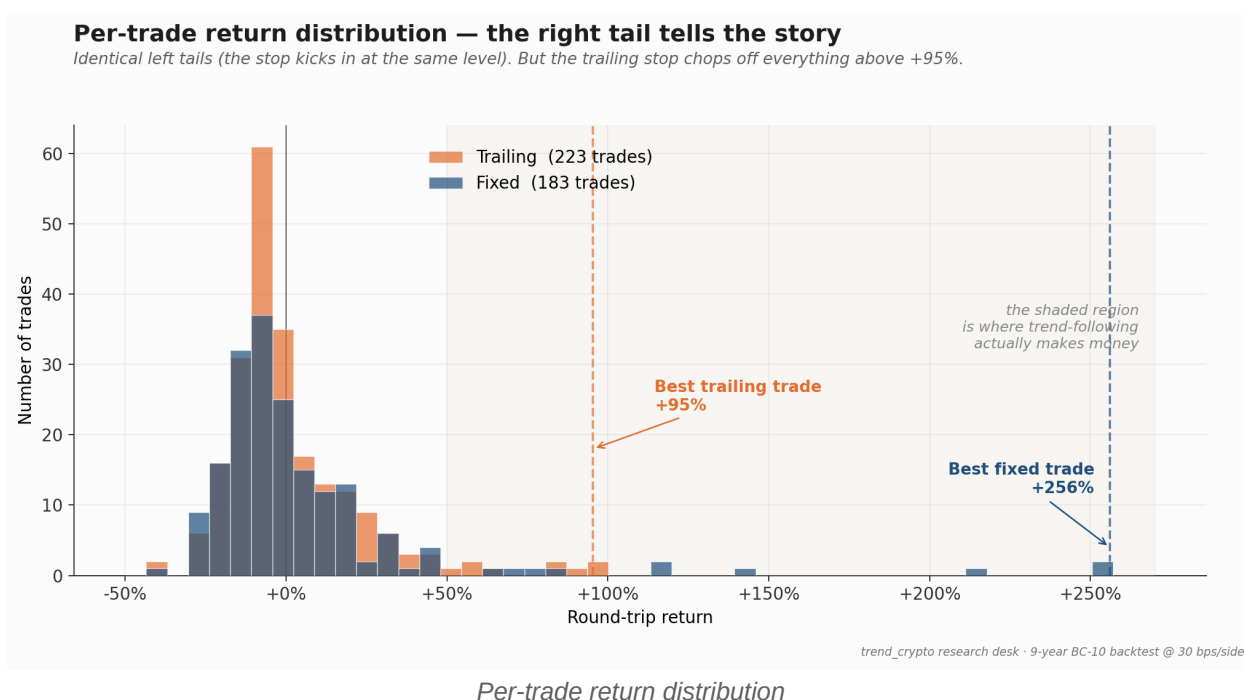
re-entered SOL at \$112.52, chasing higher, it caught only the tail of the move and gave most of that back to noise.

Same signal. Same asset. Same nine-week window. One variant captured a 4× move. The other captured roughly half of it and then paid slippage and re-entry risk to lock in less.

7. Aggregate mechanical evidence

The SOL trade is one example. Three patterns show up across the full nine-year backtest that explain the headline result.

7.1 The trailing stop chops off the right tail



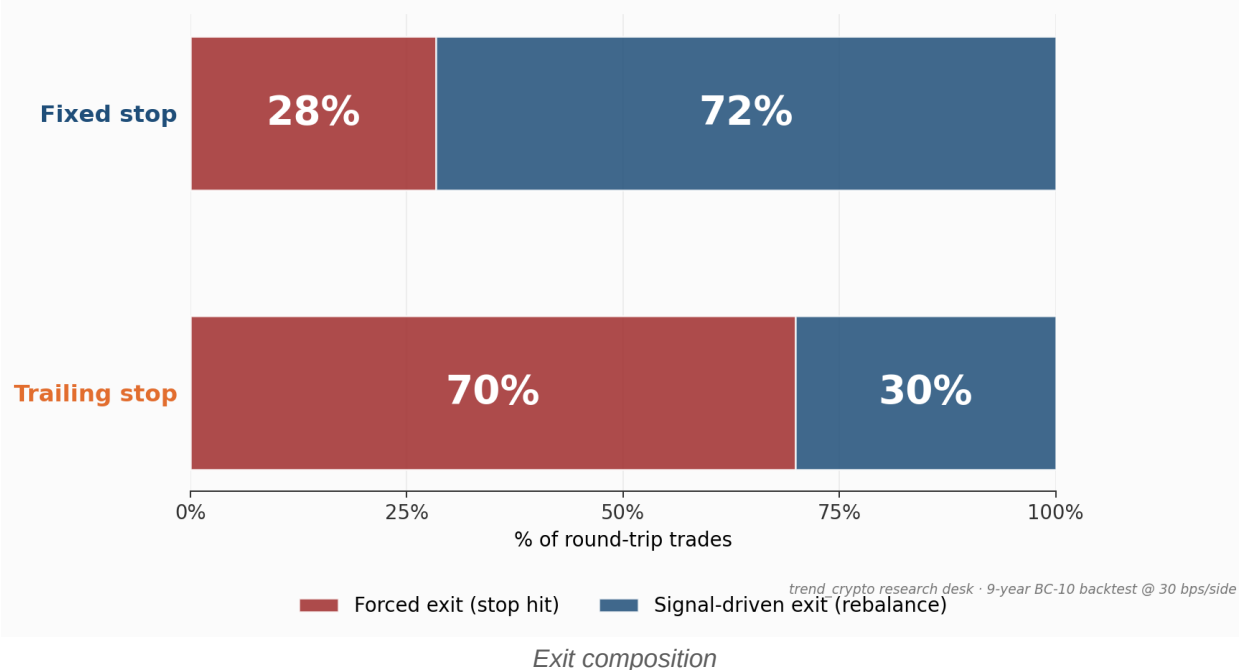
Two features of the distributions matter. First, the left tails are **identical**: losing trades are the same size in either configuration, because both stops were positioned at entry – 3×ATR at the moment of entry. The trailing stop cannot protect more on the downside than the fixed stop does — they start at the same level.

Second, the right tail is cut in half. The fixed-stop variant has a best trade of +256% and a clear cluster of trades above +100%. The trailing-stop variant has a best trade of +95% and almost nothing in the +100% bucket. The right tail is exactly where trend-following expectancy lives. Anything that systematically clips it does disproportionate damage even when median trades and left tails are unchanged.

7.2 The trailing stop displaces the rebalance signal

How each variant actually exits — and the geometry shift it reveals

Fixed: stop is the emergency brake (28%), rebalance is the edge (72%). Trailing: stop becomes the primary exit logic (70%).



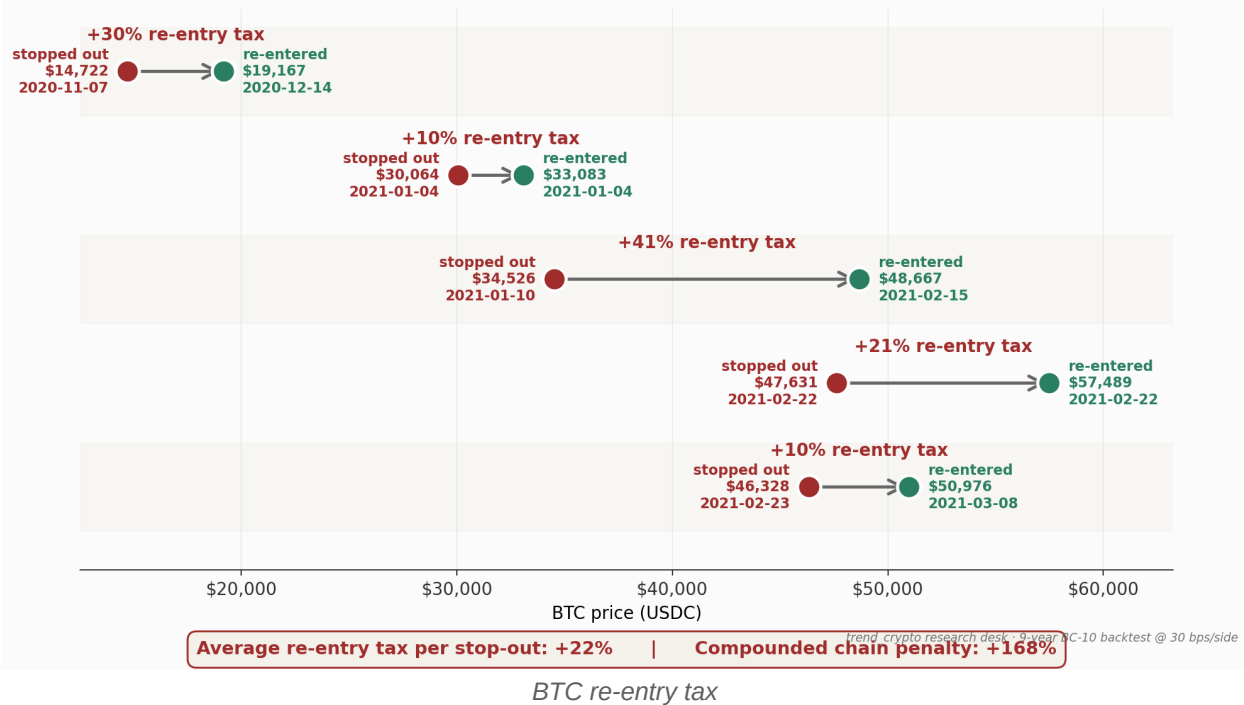
The fixed-stop variant exits 72% of trades via its rebalance signal — the strategy's actual edge — and only 28% via a stop hit. The stop is the emergency brake.

The trailing-stop variant inverts this. **70% of trades exit at the stop**, 30% on signal. The stop has become the primary exit logic. The original strategy has been replaced by a different strategy with the same entry rule. Whatever risk-adjusted edge the underlying signal contains is being filtered through a stop machinery designed around a much shorter holding horizon than the strategy actually targets.

7.3 The trailing stop forces worse re-entry prices

The trailing stop didn't just exit early — it forced worse re-entry prices

Each stop-out was followed by a re-entry at a meaningfully higher price. Five round trips, five higher fills, one persistent uptrend.



This is the part most discussions of trailing stops omit. Once a stop fires in a persistent uptrend, the only way back into the trade is a fresh upside breakout — which by construction prints at a *higher* price than the level where the trailing stop just fired.

The BTC sequence from §1, plotted as exit-prices on the left versus re-entry prices on the right, shows the pattern with no ambiguity. Average re-entry tax across the five round trips is +22% per cycle; the compounded chain penalty across the five legs is +168% — meaning the strategy re-entered at prices 168% above where the chain started, on the same underlying trend, in five months.

This is the deepest reason the trailing variant kills the strategy. It does not just clip the right tail and shift the exit mix. It converts a trend-following strategy into a churn engine that systematically buys high and sells slightly-less-high across the same uptrend it was designed to capture.

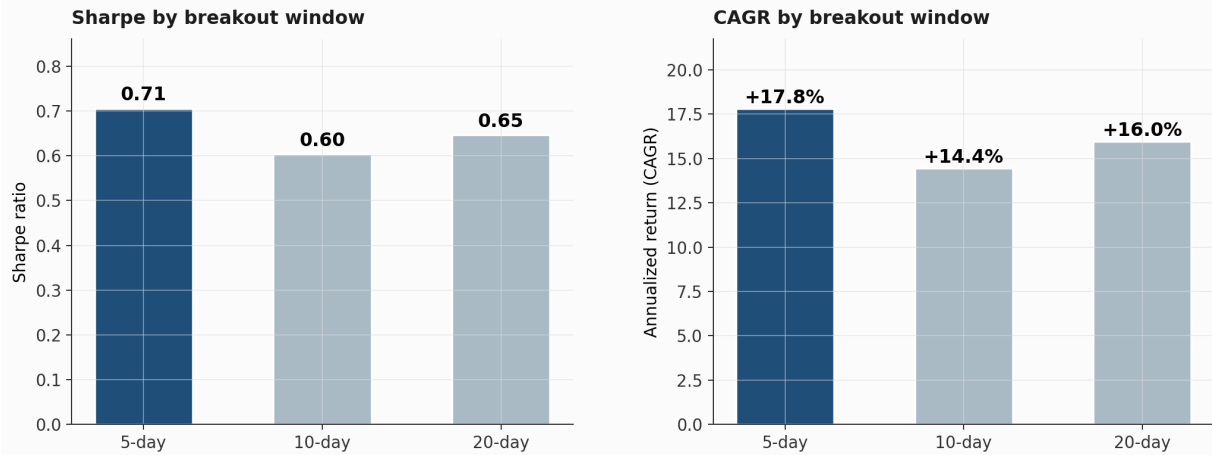
8. Robustness

8.1 Breakout window

A reasonable question is whether the 5-day breakout filter is too tight and whether the result depends on it. We re-ran the fixed-stop variant with breakout windows of 5, 10, and 20 days, holding everything else constant.

Robustness — breakout window

Same fixed 3×ATR stop, only the entry filter changes · spec value (5 days) dominates the wider alternatives



Crypto trends move quickly. Wider breakout windows enter later, give up the first leg of the move, and degrade both risk-adjusted and absolute return.

trend_crypto research desk · 9-year BC-10 backtest @ 30 bps/side

Breakout-window robustness

Breakout window	Sharpe	CAGR	Max DD	Total return
5-day	0.71	+17.8%	-52.5%	+339%
10-day	0.60	+14.5%	-54.3%	+238%
20-day	0.65	+16.0%	-54.2%	+281%

The 5-day spec dominates on both Sharpe and absolute return. Crypto trends move quickly; wider windows enter later, give up the first leg of the move, and reduce the proportion of the trend captured. The spec we analyzed in §1–§7 is not a tuned outlier — it is the strongest member of the natural neighborhood. This is a meaningful robustness check because if the result *had* depended on a precisely-chosen breakout window, the stop comparison would have been less informative.

8.2 The no-stop variant as the upper bound

Within the strategy family, the no-stop variant is the appropriate upper bound for what the entry signal can deliver before any drawdown machinery is applied. The fact that no-stop only beats the fixed-stop variant by 150 bps of annualized return (Sharpe 0.74 vs 0.71) is what makes the fixed-stop trade defensible: drawdown protection is being bought at a reasonable price. The fact that the trailing variant underperforms the no-stop variant by **1,628 bps** of annualized return for an additional 2.1 percentage points of drawdown protection vs fixed is what makes the trailing trade indefensible.

9. When trailing stops are warranted

The trailing stop is not a stupid construct in the abstract. It is wrong for this problem and right for others. Trailing stops make structural sense in two regimes:

- **Short-horizon momentum.** If the entry is a 1-3 day signal expecting the move to last 5-10 days, a tight trailing stop is appropriate. The alpha lives in small consistent moves rather than rare multi-month trends. You actively *do not* want to ride through a 25% pullback, because that pullback is a large fraction of your expected total move. Stop distance should be small relative to the trend being captured.
- **Markets with smooth, monotonic trends.** Institutional FX trends often unfold with low intraday volatility. Some commodity trends behave similarly. In those environments a 3×ATR cushion above the recent high is a reasonable "this trend is broken" signal, because the markets themselves do not typically pull back 15%+ during healthy trends.

Crypto bluechip trend-following has neither property. Trades are designed to last weeks to months. Crypto bull markets routinely have 20-30% pullbacks. The trailing stop's geometry is wrong for the volatility-to-trend ratio it is being applied to.

The general principle worth taking away: **the right stop distance is a function of the volatility-to-trend ratio of the moves the strategy is trying to capture, not a universal rule.** When trying to capture moves much larger than typical pullbacks, the stop should widen as the trade succeeds. When trying to capture moves comparable to typical pullbacks, the stop should follow along. Imposing the wrong geometry on the wrong problem — because the wrong geometry sounds cleverer in the abstract — costs measurable money.

10. Conclusion

The headline result is a clean monotone ordering of CAGR (None > Fixed > Trailing) against an almost flat ordering of max drawdown (None < Fixed < Trailing, separated by only 5.5 percentage points). The fixed stop buys drawdown protection at 44 basis points per percentage point of reduction. The trailing stop buys it at 296 basis points — almost seven times as expensive. The cost difference is not statistical noise: it is mechanical, and it follows directly from how the two stops behave as a trend runs.

Two implications worth carrying away from this. The first is specific: practitioners running trend-following strategies on crypto with weekly or longer holding horizons should consider treating the trailing stop as a contraindicated design choice for this problem rather than as a generic upgrade. The second is more general: plausible-sounding rules deserve no benefit of the doubt against backtest data, especially when the rule's premise depends on an implicit model of market behavior that the data can directly contradict. Trailing stops "lock in gains." That is the premise. Locking in gains only matters if those gains would otherwise be lost. The empirical question — would they have been lost? — is directly answerable, and the answer for crypto bluechip trend-following over this window is no: on average, the gains the trailing stop locks in are gains the trade would have kept anyway, while the cost of locking them in (forced exits, chased re-entries at higher prices, multiplied transaction costs, a

clipped right tail, an inverted exit mix) is large.

We ran the experiment because the question seemed genuinely open. We recommend that any team running a trend strategy run the equivalent experiment on its own signal. The backtest is straightforward. If the fixed-stop variant beats the trailing variant by more than 30 bps of Sharpe, the geometry of the underlying market's pullbacks is probably similar to what is in our data. If the trailing variant wins, that is also useful information — the edge may be shorter-horizon than assumed, and sizing, holding period, and re-entry assumptions should be re-examined to match.

The honest work is figuring out which kind of edge the strategy actually has, and matching the stop geometry to the shape of the trend it is trying to capture.

This piece is one of a series on a weekly-rebalanced crypto trend-following strategy across the Coinbase USDC bluechip universe. The backtest engine, parameter sweeps, and diagnostic scripts that produced every chart above are available on request. Nine years of daily data; 30 bps per side cost assumption; 100% long-only gross exposure.

*Next in the series: a rebalance-frequency study (daily vs weekly vs biweekly vs monthly), and why walk-forward parameter optimization made our Sharpe **worse** — and what that says about the way most quant strategies are tuned.*